



NITI Aayog



SECTOR AT A GLANCE 04

Flood Management in India

Water in India

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FLOOD MANAGEMENT IN INDIA

WATER IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

The Indian subcontinent, due to its topography, location, and climactic conditions, has been deeply affected by floods for many millennia. The historical record has a constant mention of floods across all periods ranging from the Harappan civilization which relied on floods for agriculture to the British era which saw devastating floods in different parts of the country.

In modern times, India witnesses a range of flooding events, including riverine floods, coastal floods, and urban floods, each year, which cause extensive damage to life, property, and the economy. Approximately 98 million Indians are exposed to extreme flood events annually.¹ Between 1980 and 2017, India experienced 278 floods, affecting over 750 million people, and causing losses worth USD 59 billion.²

Climate change is increasing the intensity and prevalence of these events and only in the last few years, extreme floods have taken place in Tamil Nadu, Bihar, Odisha, Assam, Uttarakhand, and several other states. Going forward, the frequency of flash floods, landslides, and heavy rainfall from cloudbursts in the Himalayan River basins is set to increase, and coastal flooding is likely to intensify as well.³

Effective flood management, covering both structural and non-structural measures, is therefore critical to reduce flood-related fatalities, lower economic losses, and improve climate resilience.⁴

Several government programs, including but not limited to the National Water Policy and Flood Management and Border Areas Programme, focus on effective flood management. These efforts, along with several state- and local-level policies especially in states such as Assam, Bihar, Odisha, and Kerala, seek to adopt an integrated flood management approach and build resilience against flood-related damages. Done well, these policies and programs can promote a well-balanced mix of structural and non-structural approaches, early warning systems, infrastructure management, as well as community engagement. In practice, however, these elements don't come together seamlessly and are often not customized to local realities and needs.

Going forward, emphasising institutional structures that enable localisation and collaboration, financial instruments that support floodplain zoning, and early warning systems that are accurate and drive effective community behaviour change will be critical to managing flooding risks for India. This note presents key trends related to different components of flood management in India and some ideas for the way forward.

Figure 1: Key flood management statistics



¹ CEEW, Preparing India for Extreme Climate Events, 2020

² ADB, Impacts of Natural Disasters on Households and Small Businesses in India, 2019

³ MoES, Assessment of Climate Change over the Region, 2021

⁴ NITI Aayog, Report of the Committee Constituted for Formulation of Strategy for Flood Management Works in Entire Country and River Management Activities and Works Related to Border Areas (2021–26), 2021

Key trends

This section dives deeper into India's flood management approach, outlining statistics and trends on:

1. Flood Forecasting
2. Flood Prevention Infrastructure
3. Floodplain Restoration
4. Land Use Planning
5. Community Awareness

Flood Forecasting

Flood forecasting involves scientific processes related to predicting and estimating the occurrence, extent, and severity of floods to facilitate timely actions for risk reduction and preparedness.⁵ It is considered one of the most effective non-structural measures to manage risks, save lives, and reduce economic damage.

India's forecasting system covers all major flood-prone rivers in the country:

The Central Water Commission operates 332 forecasting stations that cover all important flood-prone interstate rivers and issues approximately 7,000 forecasts from these with an accuracy rate of 95.34%.⁶

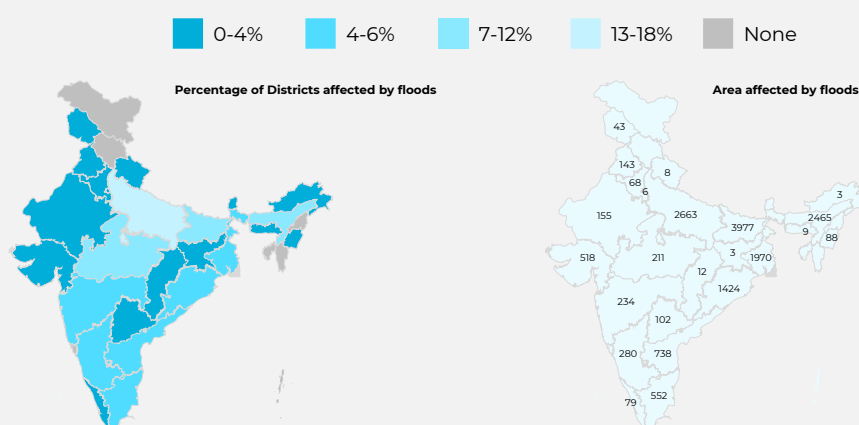
There is uneven deployment of forecasting mechanisms across Indian states:

Some states like Assam and Bihar have 20% (73) of all the forecasting stations in the country, given their vulnerability to floods. However, there are still flood-prone states like Kerala and Uttarakhand with only 3% of all stations and have considerable scope for broadening this coverage to avoid devastating floods.⁷

The accuracy of inflow flood forecasts needs to be enhanced:

Inflow forecasts play a pivotal role in flood prevention by providing crucial information on the expected volume and timing of water inflow into rivers, reservoirs, and other water bodies during heavy rainfall events. The accuracy of inflow forecasts has fallen from about 97% to 88% (from 2011-20)⁸, and improving this will be critical going forward.

Figure 2: Share of flood-affected districts (1998-2022) and flood-affected area (in hectares) by state



5 Ministry of Jal Shakti, Flood Forecasting and Early Warning System, 2022

6 NITI Aayog, Report of the Committee Constituted for Formulation of Strategy for Flood Management Works in Entire Country and River Management Activities and Works Related to Border Areas (2021–26), 2021; Ministry of Jal Shakti, Flood Forecasting and Early Warning System, 2022

7 Standing Committee on Water Resources, Flood Management in the Country including International Water Treaties in the field of Water Resource Management with particular reference to Treaty/Agreement entered into with China, Pakistan and Bhutan, 2021

8 Ministry of Earth Sciences, Technology of Forecasting of Floods, 2021; CAG, Schemes for Flood Control and Flood Forecasting, 2017

Flood Prevention Infrastructure

India has traditionally emphasised the implementation of structural measures, i.e., measures focused on the development or upgradation of infrastructure to reduce hazard risk for flood management. Flood prevention infrastructure includes dams and reservoirs, embankments, artificial river channels to control river flooding, and stormwater drains to prevent urban flooding.

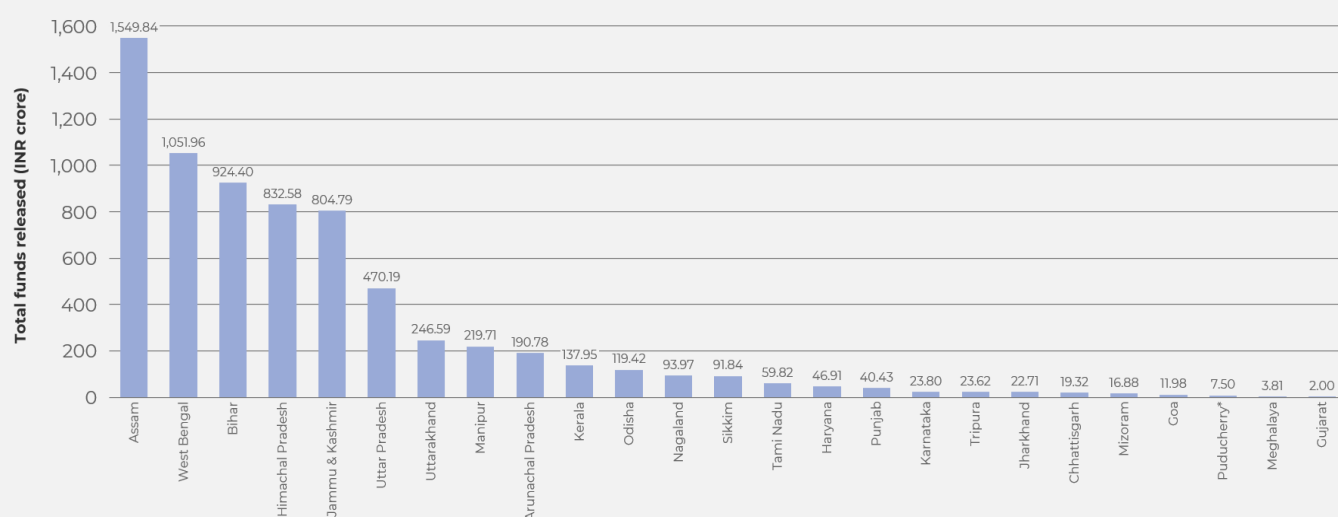
India has built an extensive network of flood control infrastructure over the years:

India has a network of 5,700 dams, 34,000 km of embankments, and 50,000 km of stormwater drainage channels covering an area of 16.5 million ha under flood protection.

However, multiple structures are ageing or are underutilised:

Around 18% of the dams in India will reach the age of 50 years in 2025, which is the average age when dams start showing signs of ageing, e.g., high sedimentation rate, and spillways.⁹ Stormwater drains are inadequately designed, with their capacity significantly lower than the rainfall they need to manage.¹⁰

Figure 3: State-wise funds released under the Flood Management Programme during the XIth and XIIth Plans and the FMABP scheme (2017-22) up to 31.03.2023



Source: Ministry of Jal Shakti, 2023

Floodplain Restoration

Floodplain restoration is the process of returning a floodplain to its condition before modification of the landscape for development or agriculture. This involves re-establishing natural floodplains, vegetation, and hydrology to mitigate flood risks and enhance ecosystem resilience.

The policy momentum on floodplain zoning needs urgent attention:

While the National Floodplains Zoning Policy helps in demarcating areas prone to flooding to regulate development in that region and help in reducing damage due to floods, only a few states have enacted this, and implementation has lagged significantly.¹¹ This is a crucial area that needs urgent attention to prevent economic damage and the loss of human life.

Early showcase projects on floodplain restoration should be scaled up:

9 United Nations University, Ageing Water Storage Infrastructure: An Emerging Global Risk, 2021

10 ICED, Climate Change Adaptation in India in the context, 2022

11 Ministry of Jal Shakti, Floodplain Zoning Policy, 2022

The Yamuna Floodplain Restoration project, which covers an area of around 1515 hectares, is one of the country's comprehensive projects focused solely on restoring floodplains.¹² It is critical to harvest insights from this and scale up similar programs across locations.

Land Use Planning

Land use planning is crucial for flood management and determines how land is developed and utilised in flood-prone areas.¹³ It involves strategically managing and zoning land to minimise flood risks, preserve natural floodplains, and promote resilient development practices.

Prohibition of development activities on riverfronts and floodplains is underway:

A draft notification for River Regulation Zones proposing to prohibit or regulate developmental activities on riverfronts and floodplains has been circulated. Such regulation for floodplains is essential to prevent illegal construction, which leads to increased risks for floods and disturbs the ecological balance.¹⁴

Community awareness

Community awareness related to flood management has seen progress in recent years, with policy efforts to increase citizen participation and programs focused on training them on preparedness, mitigation, rescue, and relief efforts during floods.

Several national policies have recognized the importance of community awareness for flood safety and risk reduction:

The National Disaster Management Authority recommends that community engagement plans be dovetailed into the Panchayat, Block, and District disaster management plans.¹⁵

Current efforts towards training community volunteers are being scaled-up:

In 2016, NDMA piloted the Aapda Mitra Scheme to train 6000 community volunteers in 30 flood-prone districts across 25 States and UTs on flood response. After a successful pilot, the NDMA approved the scale-up of the scheme to 350 districts to train 1,00,000 community volunteers by 2023.¹⁶

¹² Delhi Development Authority, Yamuna Riverfront, accessed in June, 2023

¹³ Associated Programme on Flood Management, The Role of Land-Use Planning in Flood Management, 2016

¹⁴ Ministry of Housing and Urban Affairs, River Centric Urban Planning Guidelines, 2021

¹⁵ National Disaster Management Authority, NDMA Guidelines – Management for Floods, 2008

¹⁶ MHA, Community Participation in Disaster Management, 2022; NDMA, Aapda Mitra Scheme, Accessed December 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective flood management practices. These revolve primarily around the potential for improved resilience against flood-related risks through the adoption of flood prevention and mitigation strategies.



Reduction in flood-related fatalities:

7.17 million hectares (mha) of land area are affected annually by floods in India.¹⁷ Between 1953 and 2018, on average, floods claimed the lives of 1,600+ humans and 600,000+ livestock each year.¹⁸ Comprehensive flood management plans, policies, and implementation can help mitigate this loss of life.¹⁹



Reduction in economic losses:

It has been estimated that between 1953 and 2018, on average, floods damaged around 1.2 million houses annually. In 2021 alone, floods caused an economic loss of USD 3.2 billion in India.²⁰ Effective flood management can help mitigate these flood-related losses in the country.



Improvement in climate resilience:

The Intergovernmental Panel on Climate Change states that every degree rise in temperature will lead to a 3% increase in precipitation, causing the intensification of extreme weather events such as cyclones and floods.²¹ As per the World Bank, India is projected to face an intensification of climate extremes, and without adequate resilience-building measures, extreme river floods might affect an additional 13 - 34 million people by the 2040s.²² Climate-resilient integrated flood management efforts can help mitigate this.

Challenges

This section highlights some of the major challenges and constraints facing flood management in India. Efforts to resolve such challenges will involve greater utilisation of flood forecasting stations, more resilient infrastructure, and community preparedness.



Sub-optimal utilisation of flood forecasting stations and real-time data:

The current method for setting up flood forecasting stations in India is largely driven by requests from state governments, but this has previously resulted in unpreparedness during floods in some states. Further, despite having real-time telemetry data available on flood forecasts, many times agencies continue to rely on manually observed data in cases where discrepancies exist between the two sources.²³

¹⁷ NITI Aayog, Flood Report, 2021

¹⁸ NITI Aayog, Flood Report, 2021

¹⁹ NITI Aayog, Flood Report, 2021

²⁰ Relief Web, State of the Climate in Asia 2021, 2022

²¹ CEEW, Mapping India's Climate Vulnerability, 2021

²² World Bank Group, Climate Risk Country Profile: India, 2021

²³ CAG, Schemes for Flood Control and Flood Forecasting, 2017



Inadequate flood prevention infrastructure monitoring and maintenance:

Several urban flooding instances have been recorded in recent years due to challenges in drainage management.²⁴ A CAG report highlights slow progress in the rehabilitation of natural drainage systems.²⁵ Additionally, a CAG audit pointed out that in 2017, only 2 states were carrying out the prescribed pre- and post-monsoon inspection of large dams, while another 3 were executing them partially.²⁶



Limited river rejuvenation and floodplain zoning efforts:

River rejuvenation programs currently focus extensively on reducing effluent discharge, and restoration of floodplains is not a vital component in these. Additionally, the government of India circulated the MBFPZ as early as 1975, recommending states follow and frame their state-level legislation to prevent the use of areas in floodplains. Three states and one union territory have formulated floodplain zoning legislation.²⁷



Challenges in community communication and behaviour change:

The existing flood forecasting system focuses primarily on predicting and communicating river water levels, which does not effectively convey the real risk of flooding. The technical information can be challenging for the public to comprehend and understand flood risks.²⁸

24 NITI Aayog, Flood Report, 2021

25 CAG, Schemes for Flood Control and Flood Forecasting, 2017

26 CAG, Schemes for Flood Control and Flood Forecasting, 2017

27 PIB, National Floodplains Zoning Policy, 2022; International Centre for Environment Audit and Sustainable Development (ICED), Causes, impacts, risk and mitigation of Urban Flood Management in India, 2022

28 IJDRR, Flood management in India: A focussed review on the current status and future challenges, 2020

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major government bodies and national-level institutions that oversee flood management.



Ministry of Jal Shakti²⁹

The Ministry aims to effectively manage water as a national resource. The Department of Water Resources, River Development and Ganga Rejuvenation (DoWR, RD & GR) under the Ministry of Jal Shakti is primarily responsible for establishing policy guidelines and programmes for the management, development, and conservation of water, including effective flood management. The ministry also houses the **National Institute of Hydrology**, which conducts research and development activities to improve flood management by collecting and analysing hydrological data and conducting flood estimation, flood forecasting, remote sensing and GIS-based studies.



National Disaster Management Authority³⁰

The National Disaster Management Authority has issued National Disaster Management Guidelines with key measures such as developing flood preparedness plans, operationalising integrated, multi-disciplinary water resources management for flood reduction and response and coordinating between agencies like IMD, CWC, Ministry of Housing and Urban Affairs, and urban local bodies.



Central Water Commission³¹

The CWC, the nodal agency for flood forecasting and early warning systems in the country, supports flood management by performing flood forecasting and hydrological observation and disseminating forecasts to states to take action based on state needs.



Indian Meteorological Department³²

The IMD supports flood management efforts by offering river sub-basin-specific predictions for rainfall, issuing alerts for heavy rainfall, and supplying information on significant rainfall at individual weather stations and daily average precipitation for river sub-basins to CWC for riverine flood forecasting purposes.

Major policies and programs

This section highlights existing policies related to flood management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.

²⁹ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

³⁰ NDMA, Guidelines on Management of Floods, 2008; NDMA, Guidelines on Management of Urban Flooding, 2010

³¹ CWC, Central Water Commission, accessed June 2023

³² IMD, Standard operating procedure hydrometeorological services in India, 2021

Flood Management and Border Areas Programme (FMBA)³³

The Flood Management Programme (which subsequently continued as a part of the FMBA programme) aimed to strengthen the structural measures of flood management. This was done by the provision of Central Assistance to States for activities related to flood management, anti-erosion, drainage development, anti-sea erosion, etc.

National Water Policy³⁴

The Policy aims to work towards flood management and includes key mandates like the existence of a master plan for flood control and management for each flood-prone basin, provision of adequate flood cushion in water storage projects, increased emphasis on non-structural measures for flood protection like such as flood forecasting and warning, and imposition of strict regulation of settlements and economic activity in flood plain zones.

³³ Ministry of Jal Shakti, Flood Management Measures, 2023

³⁴ NITI Aayog, Report of the Committee Constituted for Formulation of Strategy for Flood Management Works in Entire Country and River Management Activities and Works Related to Border Areas, 2021

Way forward

India is likely to witness increasing floods going forward both due to climate change but also due to increasing population pressures, especially in urban areas. These flooding events are likely to be heterogenous ranging from short-term spells of flooded roads to longer-term submerging of large areas. These incidents will have significant consequences in terms of human and animal life, disease, food, and housing shortages, and economic and livelihood impacts for millions.

Indian policies and programs have components that bring together the necessary complement of elements to manage flooding – structural and non-structural approaches, monitoring and maintenance of infrastructure, early warning, and community engagement – but the challenge has been one of effective implementation grounded in local needs and realities. Some ideas that can enable progress include:

 Strengthening nodal agencies	• Effective nodal agencies can help translate national guidelines and state policies into clear and actionable programs tailored to state realities. • Supporting decentralised and contextualised policy action can help address specific vulnerabilities and flood patterns of each region, preventing one-size-fits-all approaches
 Creating multi-stakeholder platforms	• Cross-cutting and collaborative platforms between stakeholders in the government, private sector, NGOs, and academia can (i) jointly develop flood mitigation and adaptation plans (ii) facilitate knowledge exchange and cross-learning among stakeholders to promote innovative solutions
 Harnessing financial instruments	• Instruments like mandatory flood insurance linked to floodplain zoning status can reduce encroachment and enhance the effectiveness of non-structural mitigation measures
 Improving early warning systems	• Invest in strengthening early warning systems by increasing their accuracy and density, especially in vulnerable areas like coastal regions and flood-prone river basins
 Behaviour change programming	• Community preparedness and engagement plans can both drive mitigation and build effective emergency behaviours and evacuation capabilities during disasters

Recommended readings

Ministry of Jal Shakti. (2023). *Flood Management Measures*.

<https://pib.gov.in/PressReleaselframePage.aspx?PRID=1947573>

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<https://ndma.gov.in/Natural-Hazards/Floods>

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